

**Syllabus and Course Description for
SURV 616 and SURVMETH 686
Statistical Methods and Machine Learning II
Winter 2025
(Last updated on 1/6/2025)**

Instructor:	Z. Tuba Suzer-Gurtekin	Meeting Time:	Mondays: 9:30 am – 12:30 pm
Email:	tsuzer@umich.edu	Office Hours:	By appointment

Class Website: **For this course, we will be using Canvas.** Details on using Canvas can be found at the website. All the course-related material will be posted on the website: JPSM students need to create U Michigan friend's accounts to access the Canvas website. If you have not received email about this, Elisabeth Schneider (dodo@umich.edu). Course videos will also be stored on the website.

Class Meetings Class meetings will be held in person. The rooms will be G300 in ISR-Perry in Ann Arbor and 1437 Lefrak in College Park. Additionally, Zoom links for JPSM students are available on the Canvas website. Class sessions will be recorded and made available on the Canvas website after each session. MPSDS students are expected to attend the class in person. Please bring a device that you can use to log into Zoom during class time.

For this course, pre-recorded lectures will be published ahead of time. Students are required to watch these recorded lectures before the class meeting and post questions in the discussion section on Canvas beforehand. These questions will count towards participation points for the final grade.

Class meetings will include:

Discussions on questions about the recorded material

Working on problems in small groups

Elaborations on concepts from the lectures

Additionally, in-person class meetings will focus on the software implementation of concepts covered in the recorded lectures.

Overview

The purpose of this class is to build on the concepts discussed in Statistical Methods and Machine Learning I by introducing methods for dealing with categorical data and classification problems. For example, we will explore methods for predicting binary outcomes and classifying cases.

Our focus will be on understanding and applying these methods, and appreciating how various perspectives can be used to approach data analysis problems. Emphasis will be placed on comprehending the techniques rather than deriving their theoretical properties. You will apply these techniques through weekly homework problems, a project, and a final exam.

While the workload is carefully planned, it can become overwhelming without proper and timely preparation. We expect high-quality submissions for all five homework assignments and the project.

Use of ChatGPT and other AI Tools

Instructors in this course may ask you to provide evidence of your learning through written work. When they do, the writing **you submit must be of your own creation**. Your writing will naturally be informed by other written work. When it is, you must be sure to acknowledge this influence through appropriate citation.

In the past, the ‘other written work’ was always created by another human. These days, artificial intelligence (AI) tools can be used to generate text, code, and images in response to a prompt. Such automatically generated text is not your work and cannot be submitted as if it is.

Work generated by AI tools like ChatGPT is a source that may inform your writing, just as books, papers, lecture notes, websites, and the results of Google searches can. As always, it’s your responsibility to assess the reliability of every source you use. In that context, you should be aware that the output of AI tools is constructed to sound plausible, and not checked by any expert. These results can often be completely invented.

You **may not submit AI-generated code or code that is based on or derived from AI-generated code** in this course. If you use AI to support your code writing, you must properly document and highlight in the comments which parts were assisted by AI.

While AI tools can help you avoid syntax errors, there is no guarantee that the generated code is correct. It is your responsibility to identify and correct errors in program logic through comprehensive, documented testing. Additionally, even if the generated code is syntactically correct, it may have significant room for improvement, particularly in terms of separation of concerns and avoiding repetition.

Your submission should meet our standards of attribution and validation.

Course Recordings

Instructor Wishing to Record Course Activity

Course lectures may be audio/video recorded and made available to other students in this course.

As part of your participation in this course, you may be recorded. If you do not wish to be recorded, please contact [tsuzer@umich.edu/dodo@umich.edu] the first week of class (or as soon as you enroll in the course, whichever is latest) to discuss alternative arrangements.

Instructor Wishing to Prohibit Student Recordings

Students are prohibited from recording/distributing any class activity without written permission from the instructor, except as necessary as part of approved accommodations for students with disabilities. Any approved recordings may only be used for the student’s own private use. Additional information regarding course recordings and privacy concerns can be found on the UM ITS Recording and Privacy Concerns webpage (<https://safecomputing.umich.edu/protect-privacy/privacy-u-m/videoconferencing/recording-privacy-concerns>).

Required Readings

[Faraway] Julian Faraway (2016). **Extending the Linear Model with R: Generalized Linear, Mixed Effect, and Nonparametric Regressions Models. Second Edition.**

[James et al.] Gareth James, Daniella Witten, Trevor Hastie, and Robert Tibshirani (2021). **An Introduction to Statistical Learning: with Applications in R. Second Edition.**

[Class Notes] Class Notes. Steve Miller and Richard Valliant.

Singer, J. D. and J. B. Willett (1993). "It's About Time: Using Discrete-Time Survival Analysis to Study Duration and the Timing of Events." *Journal of Educational and Behavioral Statistics* 18(2): 155-195.

Ioannidis, J. P. A. (2005). "Why Most Published Research Findings Are False." *PLoS Med* 2(8): e124.

Simmons, J. P., L. D. Nelson and U. Simonsohn (2011). "False-positive psychology undisclosed flexibility in data collection and analysis allows presenting anything as significant." *Psychological science* 22(11): 1359-1366.

Buskirk, T. D., A. Kirchner, A. Eck and C. S. Signorino (2018). "An Introduction to Machine Learning Methods for Survey Researchers." *Survey Practice* 11(1): 2718.

Zhan, T. (2022). "DI 101: Basic Introduction to Deep Learning with Its Application in Biomedical Related Fields." *Statistics in Medicine* 41(26): 5365-5378.

Other References (Optional)

An Introduction to Categorical Data Analysis (2007) by Alan Agresti, John Wiley & Sons.

Deep Learning with R (2018) by Francois Chollet and J. J. Allaire. Manning Publishers.

Data Analysis Using Regression and Multi-level/Hierarchical Modeling (2007) by Andrew Gelman and Jennifer Hill.

Applied Multivariate Statistical Analysis (2002) by Richard A. Johnson and Dean W. Wichern, Prentice Hall.

Statistical Methods, 8th ed. Snedecor and Cochran.

Grading

Homework	50%
Paper	20%
Final Exam	20%
Participation (Discussion)	10%

Each assignment will be will receive a numerical score, which corresponds to the following letter grades:

A+	[98,100]
A	[93,98)
A-	[90,93)
B+	[87,90)
B	[83,87)
B-	[80,83)
C+	[77,80)
C	[73,77)
C-	[70,73)
D+	[67,70)
D	[63,67)
D-	[60,63)
F	< 60

Homework and Project Policy

- a. This course will emphasize the use of R for statistical analysis. However, with prior agreement of the instructor, it may be acceptable to use other software.
- b. For homework and the project, submit your code and output. For the calculations, students have to show all the calculations from each step.
- c. Submissions should be a PDF file. You can submit scans of your handwriting work and/or submissions can be prepared using one of the following software packages:
 - a. MS-Word format. Use MathType or other math symbol typing software for mathematical expressions.
 - b. Latex
- d. Submit via the Canvas website.
- e. Homework assignments are due **BEFORE** the class starts. Class begins at 9:30 am on Monday. Homework that is due that day should be submitted **BEFORE 9:30am**. Late homework assignments will be accepted until one week after they are assigned but will receive 50% of the points as outlined above.
- f. Late Paper will not be accepted without the prior permission of the instructor. Prior approval must be obtained at least **3 weeks** in advance.
- g. The final course grade is determined by assigning the letter grade corresponding to the result of the weighted average of the numerical scores for each of the assignments using the weights given above.
- h. Graded assignments will be returned via the Canvas website approximately a week after the due date.
- i. Students are encouraged to work together on assignments but in a productive way.
 - Limit what is shared.
 - Copying other's work will **NOT** benefit your education.

Academic Conduct

University of Michigan (UMICH)

The University of Michigan community thrives when its members treat one another with honesty, fairness, respect, and trust. The university promotes personal responsibility and integrity while prohibiting all forms of academic dishonesty and misconduct. All cases of academic misconduct are referred to the Rackham Graduate School. Being found responsible for academic misconduct typically results in a grade sanction and may involve additional sanctions from the university.

University of Maryland (UMD)

The University of Maryland, College Park, is an academic community dedicated to the pursuit of knowledge. This mission is best fulfilled when community members uphold clearly established goals and values, rooted in the principles of truth and academic honesty. All cases of academic misconduct are referred to the Office of Student Conduct. Students found responsible for academic misconduct typically receive a grade sanction and may face additional consequences, such as probation, suspension, or expulsion from the university.

The university has a nationally recognized *Honor Code*, administered by the Student Honor Council, which includes an Honor Pledge approved by the University Senate. The University of Maryland Honor Pledge reads:

"I pledge on my honor that I have not given or received any unauthorized assistance on this assignment/examination."

In this class, we will follow UMD's policy: unless specifically advised otherwise, this pledge should be handwritten and signed on the front cover of all papers, projects, or academic assignments submitted for evaluation. Students who do not write and sign the pledge will be required to confer with the instructor.

Please familiarize yourself with the respective campus policies on academic integrity:

UMICH: <https://rackham.umich.edu/academic-policies/section8/>

UMD: <https://policies.umd.edu/academic-affairs/university-of-maryland-code-of-academic-integrity>

Understanding and adhering to these policies are essential to maintaining the integrity and academic excellence of our universities.

Accommodations for Students with Disabilities

UMD: In order to receive services you must contact the Disability Support Services (DSS) office to register in person for services. Please call the office to set up an appointment to register with a counselor. Contact the Accessibility & Disability Service; <http://www.counseling.umd.edu/ads/>

UMICH: If you think you need an accommodation for a disability, please contact Services for Students with Disabilities (SSD) office to help us determine appropriate academic accommodations; <http://ssd.umich.edu>. Any information you provide is private and confidential and will be treated as such.

Student Well-Being

Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.

If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers many resources, some of which are listed at [Resources for Student Well-being](#) on the Well-being for U-M Students website. You can also search for additional resources on that website.

Sexual Misconduct Policy

Title IX prohibits discrimination on the basis of sex, which includes sexual misconduct—including harassment, domestic and dating violence, sexual assault, and stalking. We understand that sexual violence can undermine students' academic success and we encourage anyone dealing with sexual misconduct to talk to someone about their experience, so they can get the support they need. Confidential support and academic advocacy can be found with the Sexual Assault Prevention and Awareness Center (SAPAC) on their 24-hour crisis line, 734.936.3333, and at sapac@umich.edu.

Alleged violations can be non-confidentially reported to the Equity, Civil Rights, and Title IX Office (ECRT) at ecrtoffice@umich.edu.

Student Parents and Caregivers

If you are a pregnant, parenting student, or primarily responsible for providing care for a loved one or family member, and you are in need of any accommodations, please let the instructor know at your earliest convenience. You may also reach out to <https://www.mcasp.org/> and CEW+ for resources and community support.

Other Accommodations

Students who expect to miss classes, examinations, or other assignments as a consequence of their religious observance shall be provided with a reasonable alternative opportunity to complete such academic responsibilities. It is the obligation of students to provide faculty with reasonable notice of the dates of religious holidays on which they will be absent.

Please visit <http://www.provost.umich.edu/calendar/> or <https://policies.umd.edu/student-affairs/university-of-maryland-policy-on-excused-absence> for the complete University policy at the respective institutions.

Outline

Canvas outlines each week's pre-recorded videos. Each video is about 45-60 minutes long. There are required readings for each week. In addition, there are several *optional* readings each week. These optional readings use the techniques that are developed in class that week. Note: All readings associated with a given class should be completed before the class meets, except for the first class.

Class	Date	Topic	Homework Due
1	January 13	Course Overview; Categorical Data Analysis	None

Required Videos

- Lecture1_Part1
- Lecture1_Part2
- Lecture1_Part3

Required Reading

- Class Notes #9

Optional Reading

- Ford, N. M. (1967). "The Advance Letter in Mail Surveys." Journal of Marketing Research 4(2): 202-204.
- Groves, R. M., M. P. Couper, S. Presser, E. Singer, R. Tourangeau, G. P. Acosta and L. Nelson (2006). "Experiments in Producing Nonresponse Bias." Public Opinion Quarterly 70(5): 720-736.
- Keeter, S., C. Kennedy, M. Dimock, J. Best and P. Craighill (2006). "Gauging the Impact of Growing Nonresponse on Estimates from a National RDD Telephone Survey." Public Opinion Quarterly 70(5): 759-779.
- Voogt, R. J. and H. Van Kempen (2002). "Nonresponse Bias and Stimulus Effects in the Dutch National Election Study." Quality and Quantity 36(4): 325-345.

January 20	NO CLASS -- University Symposia, Martin Luther King Jr Day
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Required Videos

- Lecture2_Part1
- Lecture2_Part2

Required Reading

- Faraway, Chapter 6;
- Class Notes #10

Optional Reading

- Turner, C. F., L. Ku, S. M. Rogers, L. D. Lindberg, J. H. Pleck and F. L. Sonenstein (1998). "Adolescent Sexual Behavior, Drug Use, and Violence: Increased Reporting with Computer Survey Technology." Science **280**(5365): 867-873.
- Groves, R. M., M. P. Couper, S. Presser, E. Singer, R. Tourangeau, G. P. Acosta and L. Nelson (2006). "Experiments in Producing Nonresponse Bias." Public Opinion Quarterly **70**(5): 720-736.
- Converse, P. D., E. W. Wolfe, X. Huang and F. L. Oswald (2008). "Response Rates for Mixed-Mode Surveys Using Mail and E-mail/Web." American Journal of Evaluation **29**(1): 99-107.

3 February 3 Epidemiology

HW2Required Videos

- Lecture3_Part1
- Lecture3_Part2

Required Reading

- Class Notes #13

Optional Reading

- Kessler, R. C., S. G. Heeringa, M. B. Stein and et al. "Thirty-day Prevalence of DSM-IV Mental Disorders among Nondeployed Soldiers in the US Army: Results from the Army Study to Assess Risk and Resilience in Servicemembers (Army STARRS)." JAMA Psychiatry.
- Leung, H. M. and L. L. Kupper (1981). "Comparisons of confidence intervals for attributable risk." Biometrics: 293-302.
- Whittemore, A. S. (1983). "Estimating attributable risk from case-control studies." American journal of epidemiology **117**(1): 76-85.

Required Videos

- Lecture4_Part1
- Lecture4_Part2

Required Reading

- Faraway, Chapter 1
- Class Notes 12

Optional Reading

- Groves, R. M., E. Singer and A. Corning (2000). "Leverage-Saliency Theory of Survey Participation: Description and an Illustration." Public Opinion Quarterly **64**(3): 299-308.
- Singer, E., J. van Hoewyk and M. P. Maher (2000). "Experiments with Incentives in Telephone Surveys." Public Opinion Quarterly **64**(2): 171-188.

5 February 17 Logistic Regression**HW4**Required Videos

- Lecture5_Part1
- Lecture5_Part2

Required Reading

- Faraway, Chapter 2

Optional Reading

- Sangster, R. L. and B. J. Meekins (2004). "Modeling the Likelihood of Interviews and Refusals: Using Call History Data to Improve Efficiency of Effort in a National RDD Survey." Proceedings of the Annual Joint Statistical Meeting, American Statistical Association. Washington, DC: American Statistical Association.
- Alho, J. M. (1990). "Adjusting for Nonresponse Bias Using Logistic Regression." Biometrika **77**(3): 617-624.

6 February 24 Generalized Linear Models**None**Required Videos

- Lecture6_Part1
- Lecture6_Part2

Required Reading

- Faraway, Chapter 8

Optional Reading

- McCulloch, C. E. (2000). "Generalized Linear Models." Journal of the American Statistical Association **95**(452): 1320-1324.
- Myers, R. H. and D. C. Montgomery (1997). "A tutorial on generalized linear models." Journal of Quality Technology **29**(3): 274-291.
- McCullagh, P. (1984). "Generalized linear models." European Journal of Operational Research **16**(3): 285-292.

March 3 NO CLASS -- MICHIGAN SPRING BREAK

7-8 March 10 Statistical and Machine Learning Models None

Required Videos

- Lecture7_Part1
- Lecture7_Part2
- Lecture8

Required Reading

- Ioannidis (2005)
- Simmons et al. (2011)
- Buskirk et al. (2018)
- James et al. Section 4.1, 4.2, 4.5

9 March 17 Classification: K-Means

Required Videos

- Lecture9

Required Reading

- James et al., Section 12.4

March 24 NO CLASS -- MARYLAND SPRING BREAK

10 March 31 Classification: Neural Networks Paper

Required Videos

- Lecture10_Part1
- Lecture10_Part2

Required Reading

- James et al., Section 2, 10.1, 10.2, 10.6, 10.7, 10.8
- Faraway, Chapter 17.
- Zhan, T. (2022). "Dl 101: Basic Introduction to Deep Learning with Its Application in Biomedical Related Fields." Statistics in Medicine **41**(26): 5365-5378.

Required Videos

- Lecture11_Part1
- Lecture11_Part2

Required Reading

- Class Notes #14
- Singer and Willett (1993)

Optional Reading

- Brick, J. M., J. Montaquila and F. Scheuren (2002). "Estimating Residency Rates for Undetermined Telephone Numbers." Public Opinion Quarterly **66**(1): 18-39.
- Groves, R. M. and S. G. Heeringa (2006). "Responsive design for household surveys: tools for actively controlling survey errors and costs." Journal of the Royal Statistical Society: Series A (Statistics in Society) **169**(3): 439-457.
- Kennickell, A. B. (1999). "Analysis of nonresponse effects in the 1995 Survey of Consumer Finances." Journal of Official Statistics. **15**: 283-304.

Required Videos

- Lecture12

Required Reading

- Class Notes #15

Optional Reading

- Etter, J.-F. O., T. V. Perneger and A. Rougemont (1996). "Does sponsorship matter in patient satisfaction surveys? A randomized trial." Medical care: 327-335.

Required Videos

- Lecture13_Part1
- Lecture13_Part2

Required Reading

- Class Notes #16

Optional Reading

- Bjorner, J. B., M. Rose, B. Gandek, A. A. Stone, D. U. Junghaenel and J. E. Ware (2014). "Difference in method of administration did not significantly impact item response: an IRT-based analysis from the Patient-Reported Outcomes Measurement Information System (PROMIS) initiative." Quality of Life Research **23**(1): 217-227.
- Davidov, E., P. Schmidt and S. H. Schwartz (2008). "Bringing Values Back in: The Adequacy of the European Social Survey to Measure Values in 20 Countries." The Public Opinion Quarterly **72**(3): 420-445.
- Klausch, T., J. J. Hox and B. Schouten (2013). "Measurement Effects of Survey Mode on the Equivalence of Attitudinal Rating Scale Questions." Sociological Methods & Research **42**(3): 227-263.

Paper

Projects are designed to allow each student some individual creativity in developing and expressing their own approach to data analysis, within the framework of the stated goals of SURV 616/SURVMETH 686. This project will be based upon a provided dataset and will involve the use of logistic regression. Students in SURV 616/SURVMETH 686 are NOT expected to use the specialized techniques, taught in other courses, that have been developed for analyzing survey data. You can treat each dataset as if it were a simple random sample. However, keep in mind that if a dataset from a complex sample were analyzed appropriately, the relationships found could be different from those that you uncover.

Although fitting of statistical or machine learning models is expected, do not overlook the need for simple descriptive analyses to better understand the variables in your data. Your project goals should be to:

- (a) Demonstrate that you know how to apply the methods covered in class in a systematic and thorough way to investigate relationships in a dataset or to explore a particular problem, and
- (b) Present your results coherently in a written report.

Reports should include tables and figures that illustrate the points you want to make and the analyses you have done. The tables and figures should be formatted suitably as if for publication.

Paper: Format

The Paper should be no more than 4 pages have the following formatting: line spacing=1.5, with Times New Roman font size 12, one-inch margins on all sides. Include page numbers at the bottom of each page.

The paper should have a short introduction, a description of the methods and data that will be used, results, and a brief discussion of the results.

Paper: Grading

The Paper will account for 20% of the grade in the class.

Grades will be based on the thoroughness of analysis, correctness of results, and the quality of the written report. Points will be deducted if reports are poorly written or poorly organized.

Final Exam

The Final Exam is an open-note and open-book quiz. Students cannot share their work or answers with other students. Students cannot search for specific answers on the internet. Students may use R or Excel or other software to perform calculations. They are designed such that they can be done by hand. The exam is timed. Students will have 2 hours to complete the exam once they start.